Knowledge flows and knowledge spillovers

Who you know, where you are, and why ideas leak

Innovation Management · A.Y. 2026/27
Prof. Grazia Sveva Ascione · Wednesday 28 October 2026

Politecnico di Torino

Today's lecture

01 **Intuition: who you know matters for innovation** people, cities and new ideas

02 **Why do cities exist?** sharing, matching, learning; tacit knowledge; externalities

03 **Knowledge spillovers** flows and spillovers, how they travel, why they matter

04 **What promotes knowledge spillovers?** Marshall–Arrow–Romer, Porter, Jacobs

05 **Tracing knowledge flows** patent citations, Jaffe et al. (1993) and their limits

01

Intuition: who you know and where you live matters for innovation



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Where do new ideas come from?

> “Innovation disproportionately happens in cities. What is it about packing people together that makes them so innovative?”

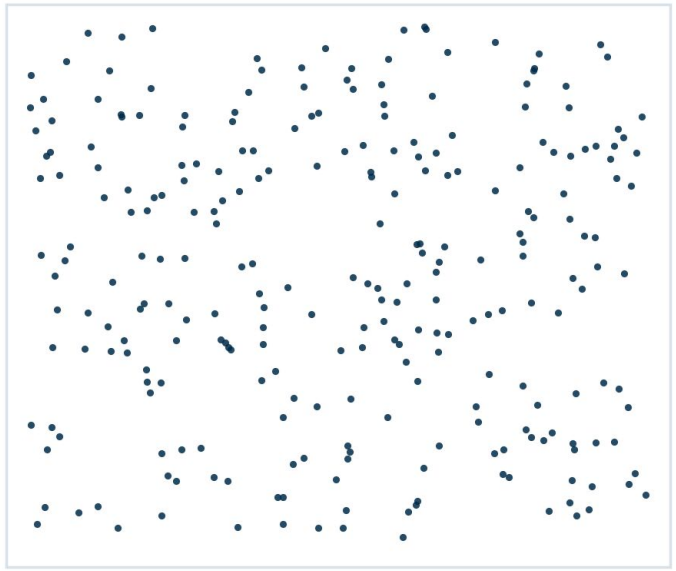
Matt Clancy (2021), Why proximity matters: who you know, New Things Under the Sun

How important is to meet with others compared to just access the knowledge they produced to create new innovation?

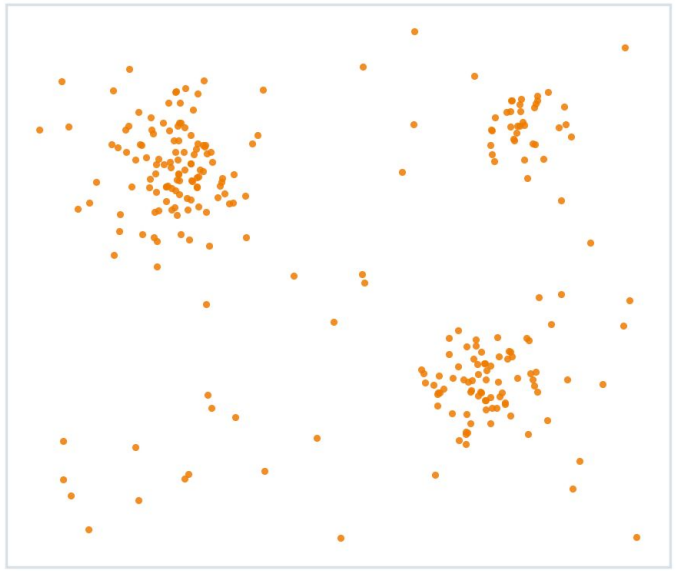
Think of the last good idea you had for a project. Did it come from something you read, or from someone you talked to? Where was that person?

Cities multiply the people you can meet

If people located at random



Where people actually live



Same number of people, same territory

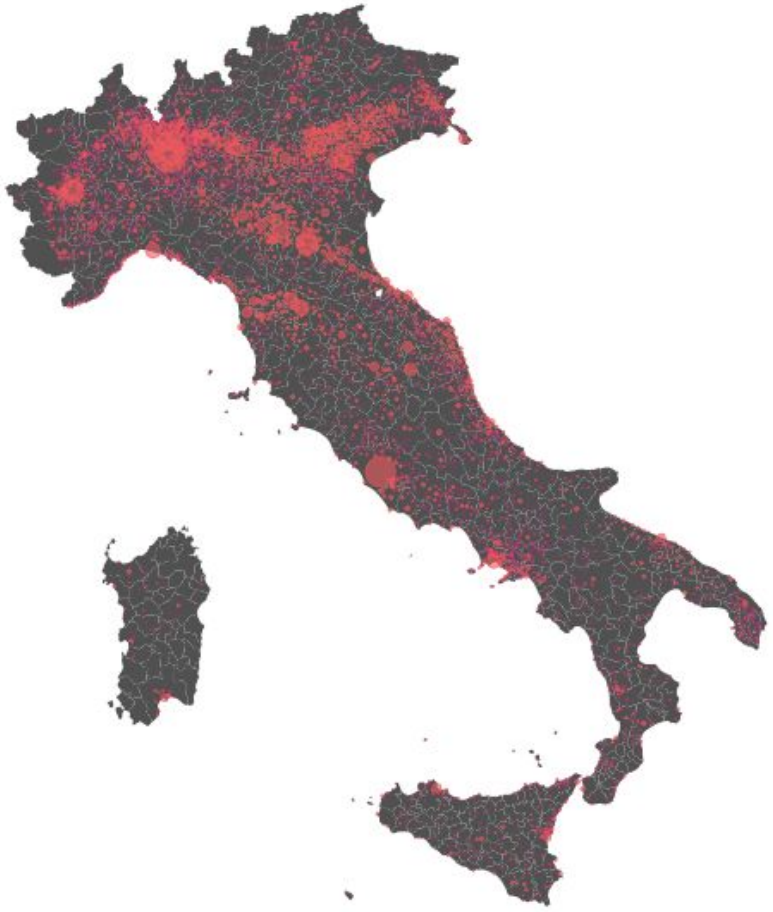
People are not spread at random: they crowd into cities, and pay higher rents and longer commutes to be there.

Innovation is even more concentrated: it happens disproportionately in cities, and in dense neighbourhoods inventors work on a more diverse set of technologies.

A city is a machine for meeting people.

> Why pay Milan rents when you could work from a village with a good internet connection?

What are the effects of proximity? More innovation



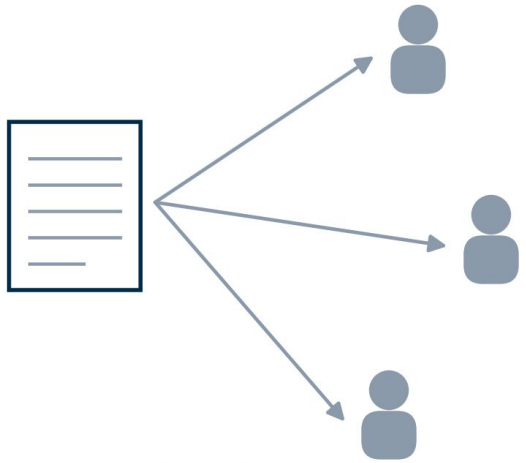
Map of patents filed in Italy in 2023 (Source:EPO)

Considering Lombardy, Emilia Romagna and Veneto, they account for ⅔ of Italian patenting but only ⅓ of the population lives there (source ISTAT)

Region	Share of Italy's Population (2024)	Share of Italian EPO Applications (2023)
Lombardy	17.0%	32.2%
Emilia-Romagna	7.6%	19.0%
Veneto	8.2%	13.5%
Total	32.8%	64.7%

Tacit knowledge vs codified knowledge

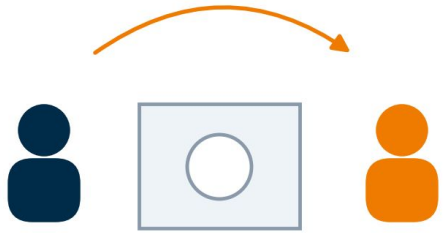
Codified: written down once



read anywhere: travels far

Tacit: shown face to face

"feel" and experience



needs proximity: stays close

Codified knowledge

can be written down, stored and sent: documents, manuals, patents, databases.

Example: a software manual, or a patent describing a process.

Tacit knowledge

is hard to write down; it is learned by experience, observation and practice.

Example: a senior engineer shows a junior how to fine-tune a machine, by "feel".

How knowledge travels: knowledge flows



Codified knowledge travels far

Patents and papers can be read anywhere. But using them requires knowing what to look for, and often someone to explain them.

Tacit knowledge stays close

Know-how lives in people: it moves when they talk, work together or change job. This is why spillovers are local, and why who you know matters.

Industrial clusters in Northern Italy: where tacit knowledge lives



What a cluster concentrates

firms, skilled workers, specialised suppliers and research institutions, in a few kilometres.

Local exchange of tacit knowledge

know-how that is hard to write down: setting up a packaging machine (Bologna), how a glaze behaves in the kiln (Sassuolo), how a ski boot must fit (Montebelluna).

How it moves

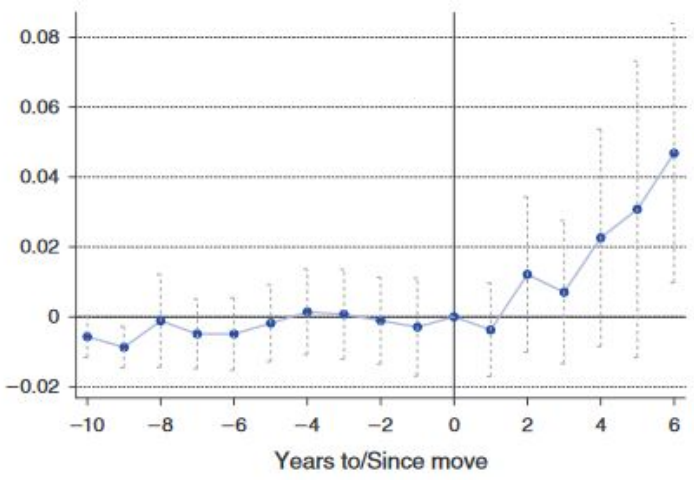
worker mobility between firms, repeated interactions with suppliers and customers, collaboration on joint projects.

The link: where knowledge is tacit, firms need to be close to learn from each other. This is why these industries cluster.

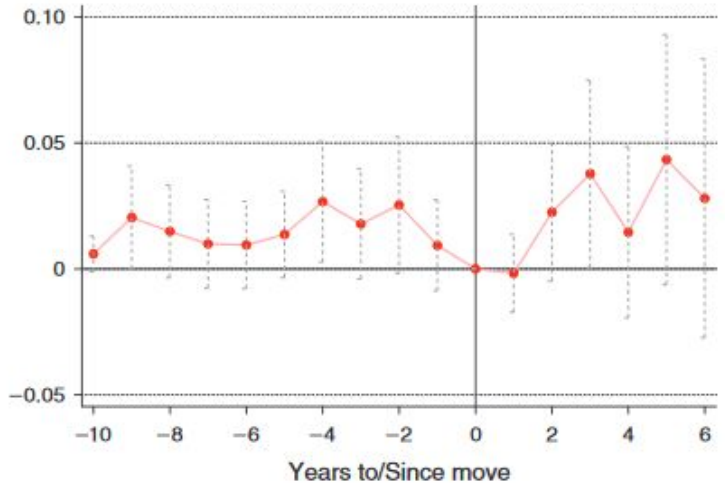
What are the effects of proximity? More collaboration in academia

The Jussieu experiment: asbestos removal at the Jussieu campus (Paris) moved labs for 17 years, wherever there was space: a natural experiment.

Labs moved into the same building
probability of collaborating, years before and after



Labs separated
probability of collaborating, years before and after



If scientists move together: joint research becomes about 3.5 times more likely.

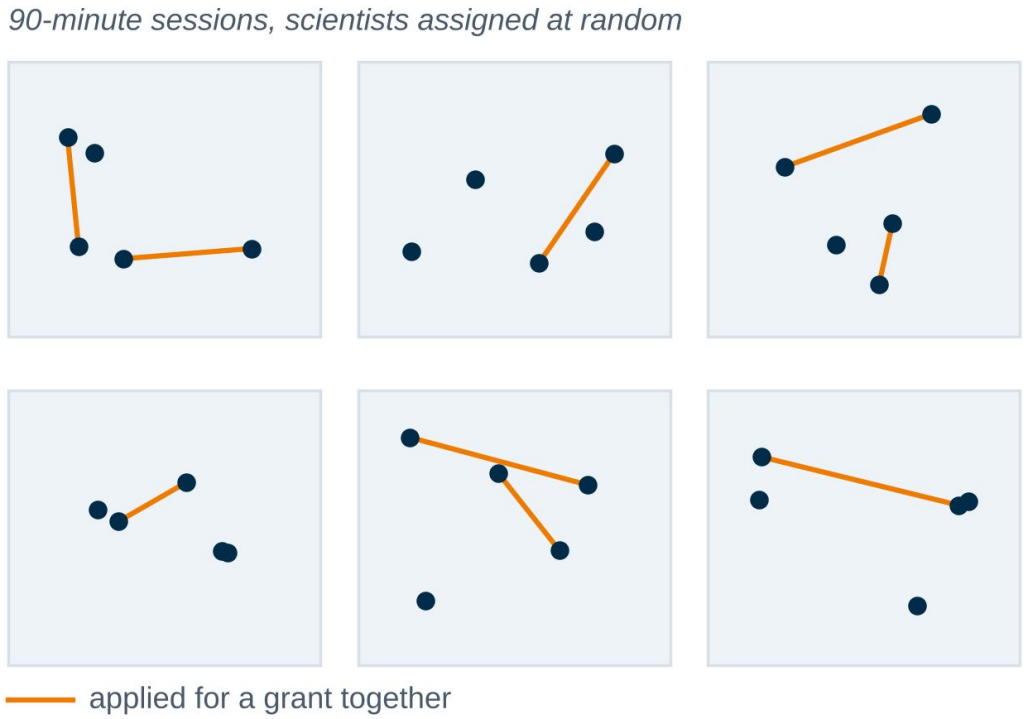
If they know each other and then separate: collaboration does not fall. Once labs know each other, distance matters less.

Different topics: the effect is largest for labs working on different things. Similar labs meet anyway, at seminars and conferences.

LOCATION MATTERS FOR INNOVATION

Catalini, C. (2018), Microgeography and the direction of inventive activity, Management Science. Figures via M. Clancy (2021).

What are the effects of proximity (I)? More interactions



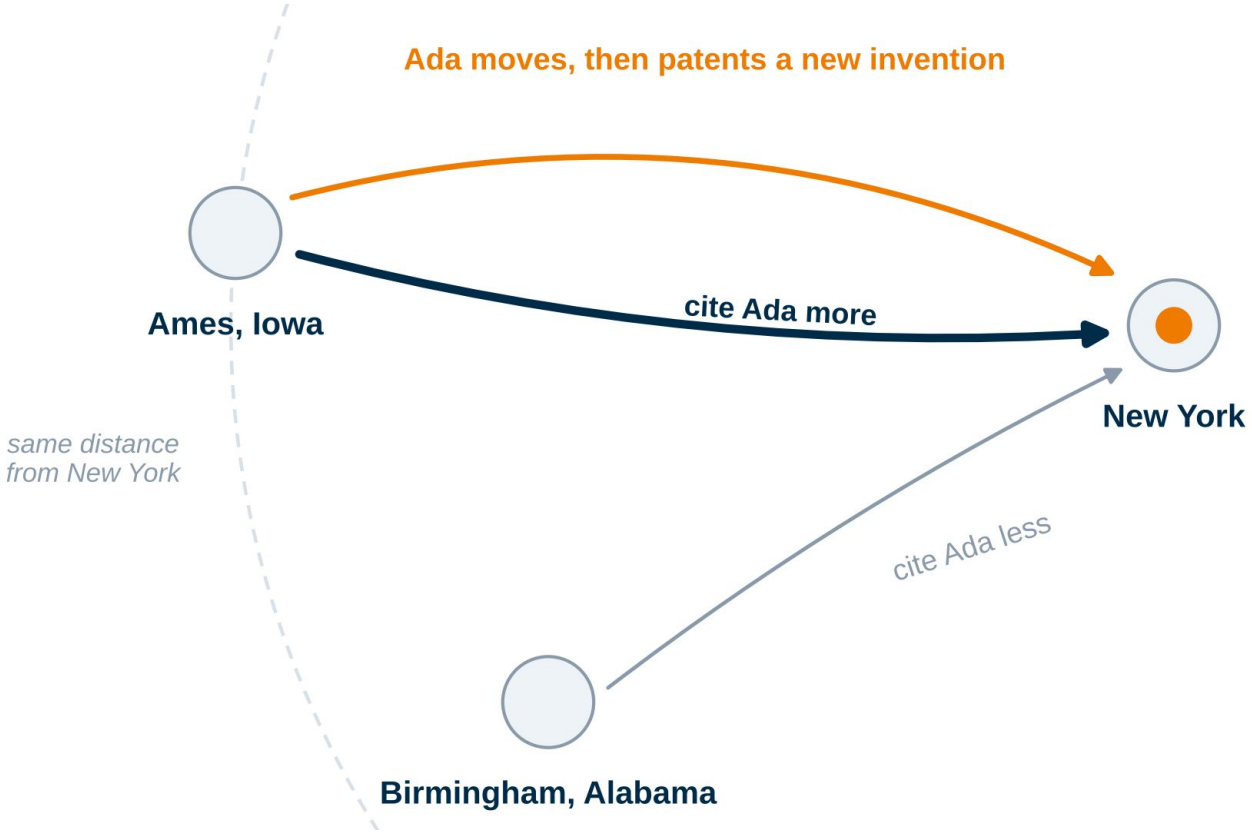
Before a grant call, Harvard scientists were assigned at random to 90-minute sessions to present their research.

Two scientists in the same session were 75% more likely to apply for the grant together.

Same institution, same corridors: still, finding the right partner is costly. Encounters lower the cost of search.

Boudreau, K. et al. (2017), A field experiment on search costs and the formation of scientific collaborations, Review of Economics and Statistics.

Does the effect of proximity last when we change location?



Ada, an inventor in Ames, moves to New York and patents a new invention.

Inventors who stayed in Ames cite it more than inventors in a city just as far from New York.

135% as many citations from Ames inventors in her field; **175%** from Ames inventors in other fields.

80% of the extra citations come from her former co-inventors and colleagues.

Message

Location is important to meet with people you would not normally meet (they don't attend same conferences, go out in same places..)

Agrawal, A., Cockburn, I. & McHale, J. (2006), Gone but not forgotten, Journal of Economic Geography.

The message: who you know shapes what you learn

> **Innovators meet other innovators who live nearby, whether they work in the same field or not.** Once a relationship is formed, it stays productive even after they move apart: they can still work together, or at least learn from each other.

Proximity creates ties

People who are close meet more, and start working together.

Seen today: Jussieu: labs moved into the same building collaborate more.

Chance meetings matter

Meeting by chance lowers the cost of finding the right partner.

Seen today: Harvard: scientists in the same session apply for grants together.

Ties survive distance

Once formed, a tie keeps carrying ideas after people move.

Seen today: Ada: her former colleagues keep citing her from New York.

Next part: why do people and firms crowd into cities in the first place?

02

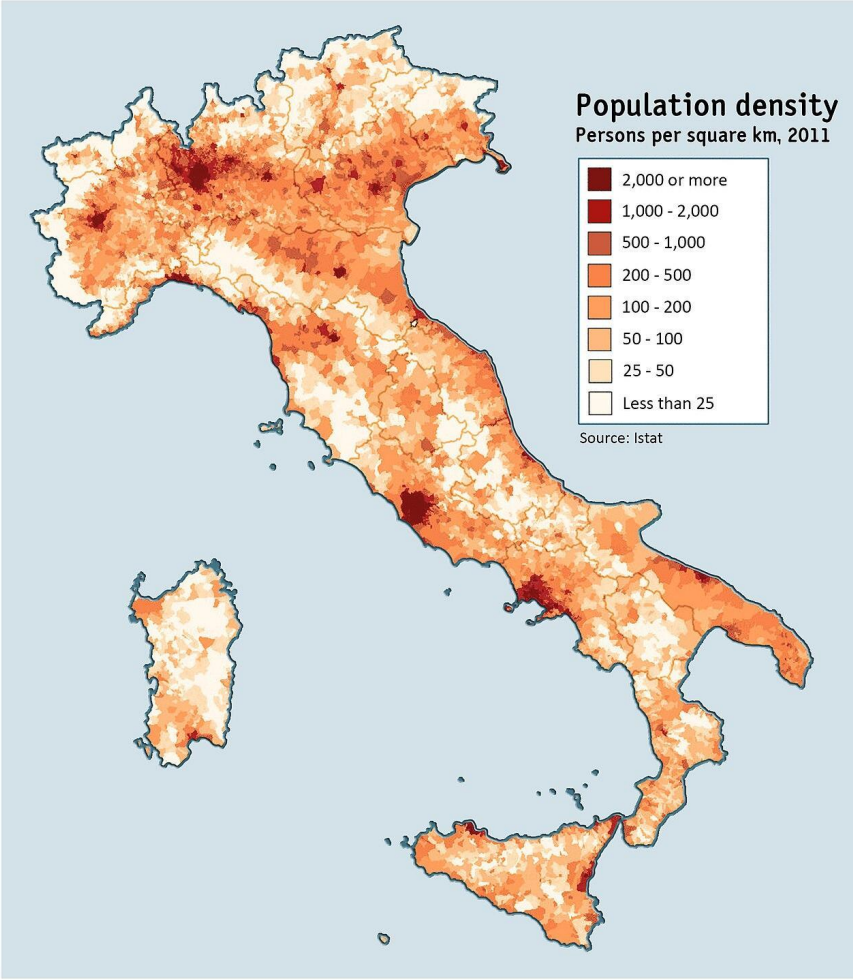
Why do cities exist? Why do we live in cities?

Sharing, matching, learning



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Do we choose where to live by chance? What drives that choice?



ISTAT, population density by municipality (persons per km²), 2011 Census.

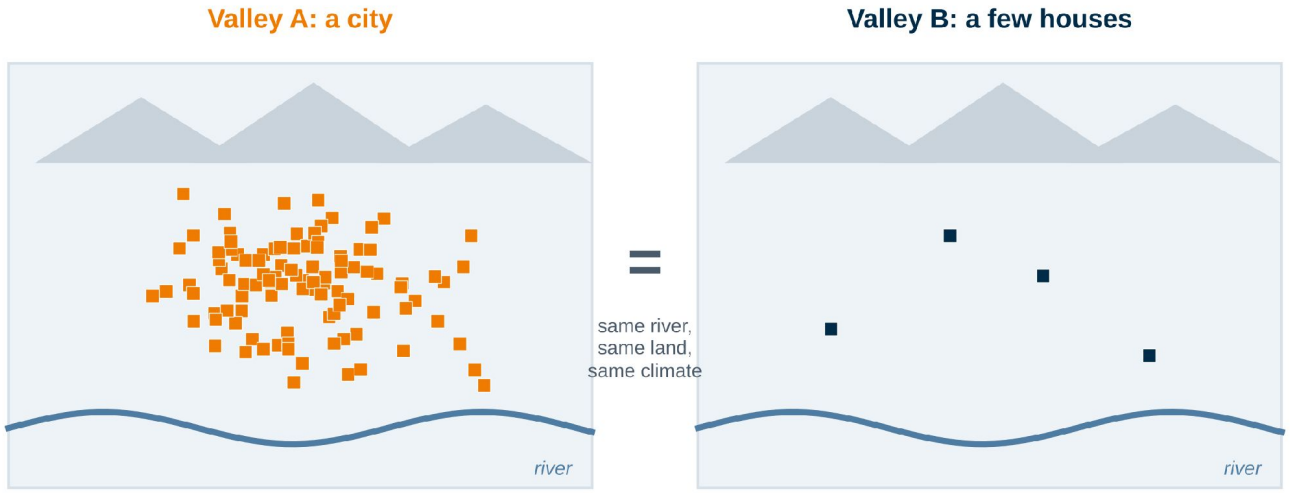
Population density in Italy, 2011

People are packed in the Po valley, along the coasts and around a few large cities; large areas are almost empty.

Question for you

Why do so many people live in Milan, Turin, Rome or Naples? List as many reasons as you can.

The economics of agglomeration: a preference for proximity beyond natural resources



Same natural resources, very different density: the difference is agglomeration

Why people are where they are

Chance: random location choices would give a uniform pattern. We do not observe it.

Natural advantages: rivers, ports, climate and resources explain part of it.

Agglomeration economies: what remains is a preference for being close to others, for proximity itself.

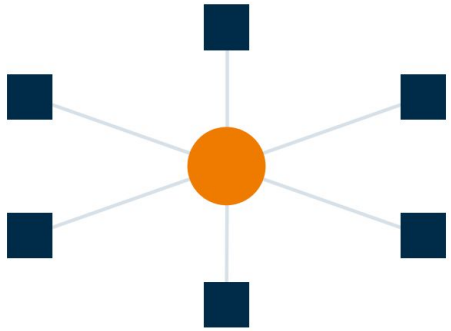
A city exists where centripetal forces prevail over centrifugal ones. Without centrifugal forces we would see one single megalopolis.

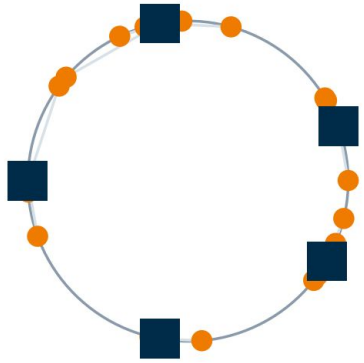
Under parity of natural resources, people and firms still concentrate. Physical geography explains only about 35% of the differences in economic activity within countries: the rest is agglomeration.

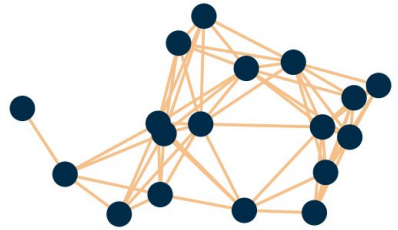
Henderson, J. V., Squires, T., Storeygard, A. & Weil, D. (2018), The global distribution of economic activity: nature, history, and the role of trade, Quarterly Journal of Economics, 133(1), 357–406.

Sharing, matching, learning: what are the benefits of proximity?

Marshall (1890) named three sources of agglomeration economies; Duranton & Puga (2004) defined the mechanisms as sharing, matching and learning.







Sharing

In a larger group it is easier to share costly, indivisible resources: specialised suppliers, infrastructure, a pool of workers.

Matching

In a larger group it is more likely to find a good match between the skills a firm needs and those workers offer.

Learning

In a larger group more knowledge is generated, spread and accumulated, and others can use it for free.

All three produce externalities: a location choice affects those nearby, without a price in between.

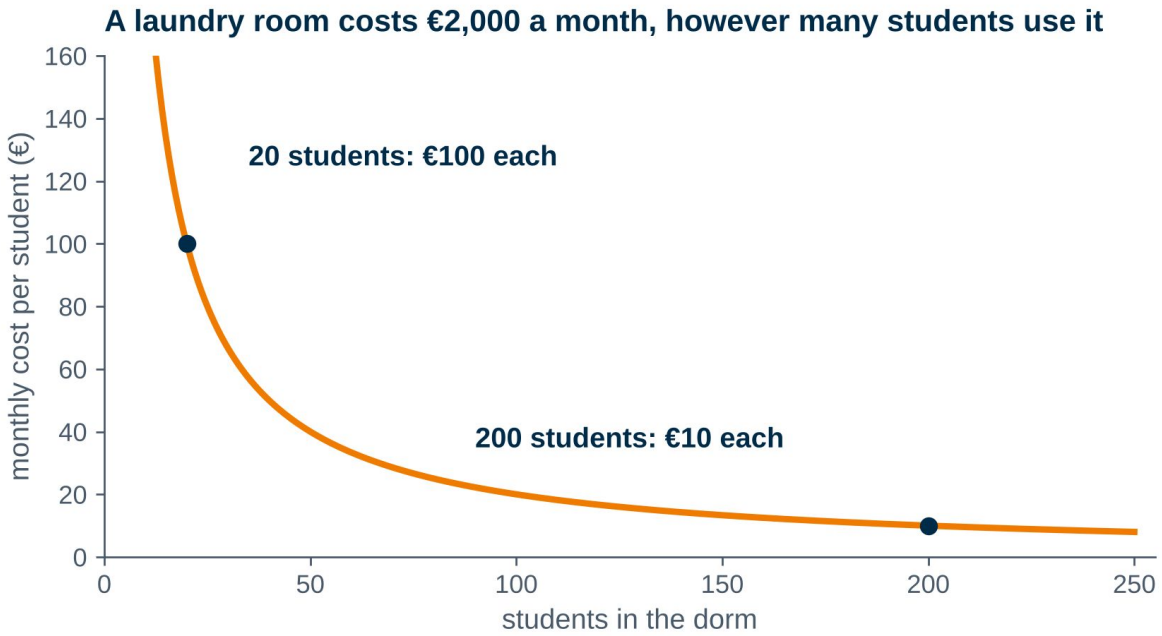
Sharing

Sharing: some useful things have high fixed costs. They become affordable only when many people or firms are in the same place and share them.

Example: a laundry room in a student dorm

The room costs €2,000 a month, however many students use it. With 20 students, each pays €100; with 200 students, each pays €10.

The cost is fixed: more users, lower price for each.



Question for you

Why does Milan have several IMAX cinemas, and Settimo Torinese none?

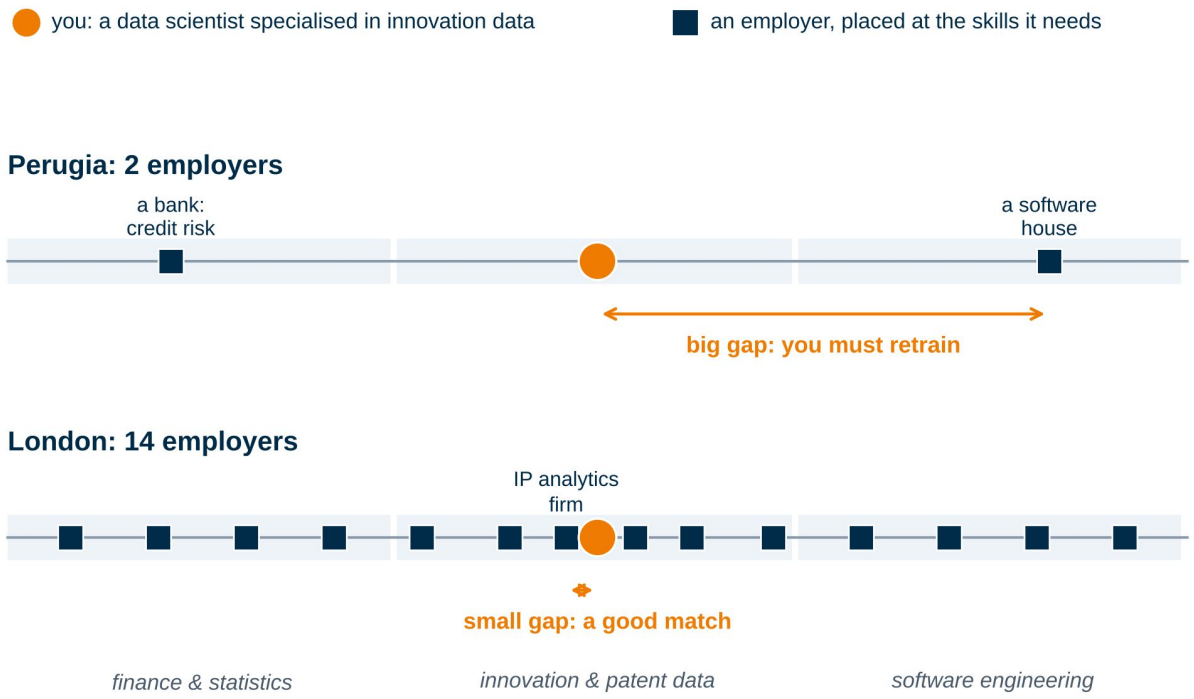
Matching

Matching: in a city workers find jobs that fit their skills, and firms find workers with the skills they need.

Example: a data scientist specialised in innovation data

In Perugia, my hometown, 2 employers need my skills; in London, 14 do.

With more employers, the closest one is closer to what she can do: a better match, more productivity, higher pay.



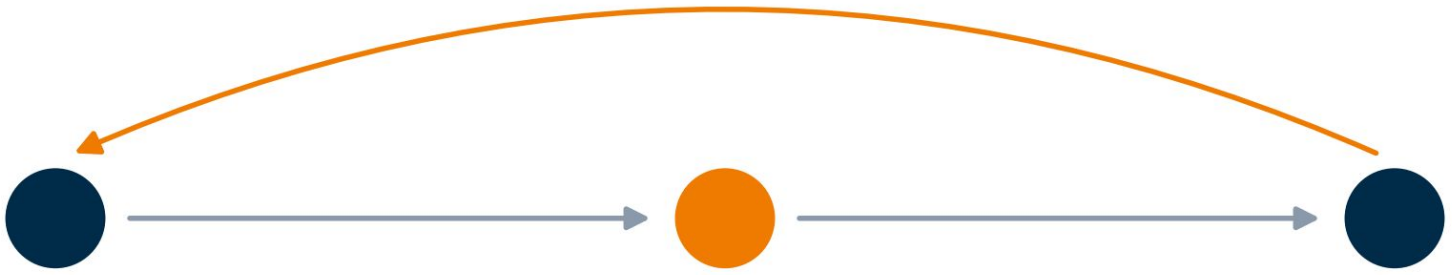
Question for you

Where would you look for a job, and why? What does the firm gain from a better match?

Learning

Learning: cities make it easier for people and firms to acquire, exchange and improve knowledge, because they meet others more often.

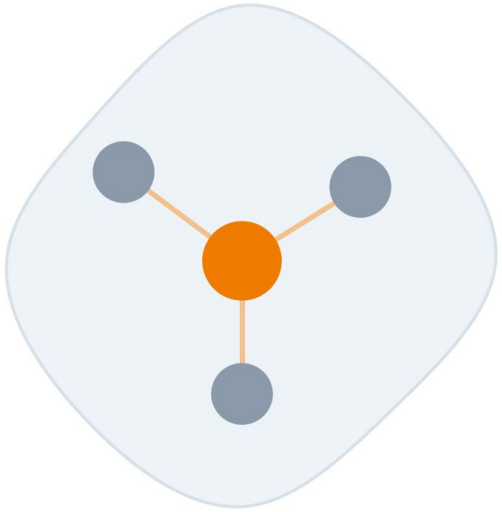
knowledge produces more knowledge (Romer, 1986)



Knowledge produces more knowledge: what is generated spreads, accumulates, and can be used by others nearby, often for free.

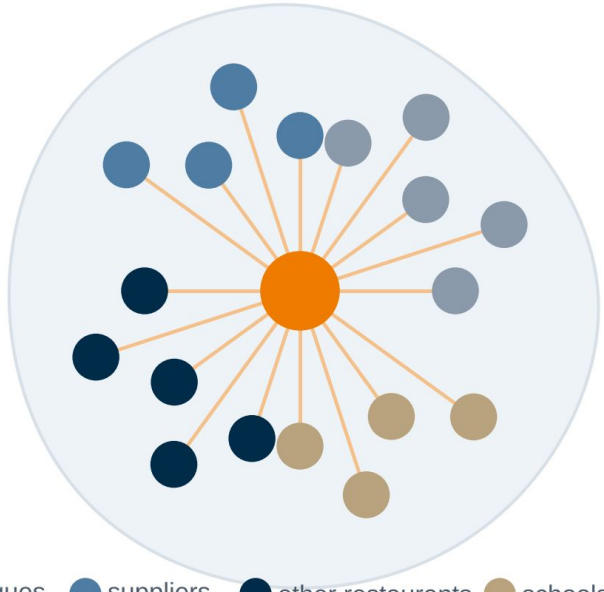
Learning: a junior chef in a village and in Milan

A village: one restaurant



3 colleagues to learn from

Milan: hundreds of restaurants



colleagues · suppliers · other restaurants · schools and events

In a village with one restaurant

The chef learns from the same few colleagues.

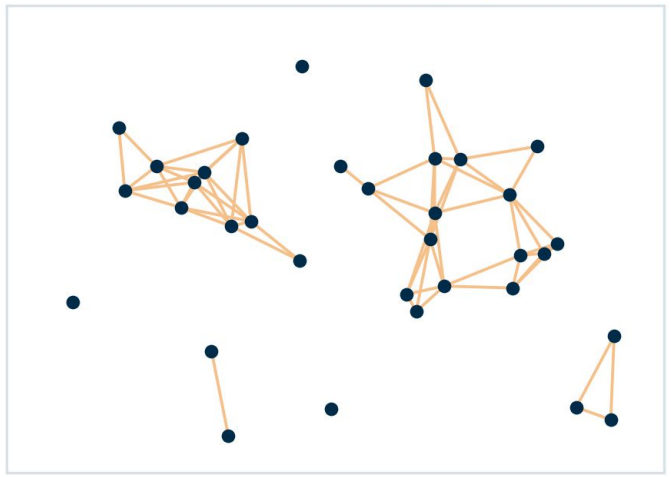
In Milan, with hundreds of restaurants

The chef learns new techniques from colleagues, new ingredients from suppliers, new ideas from other restaurants, new practices when she changes job, new trends at schools and events.

Small place: few interactions, slow learning. **Large city:** many interactions, fast learning. The same holds for firms.

Diffusion of knowledge is a matter of density

Dispersed — 34 people · 66 links



In a city — 34 people · 139 links



A link = two people close enough to meet regularly

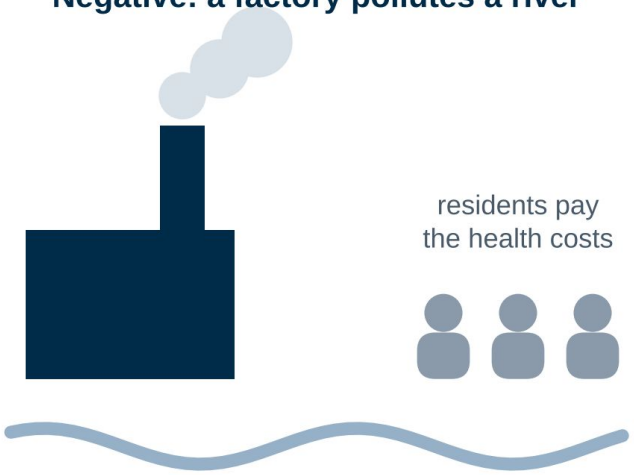
Knowledge spreads through **networks of people**: **more people and more links mean faster diffusion**.

Same people, same rule for a link: concentrated in a city, they have twice as many links.

What is an externality?

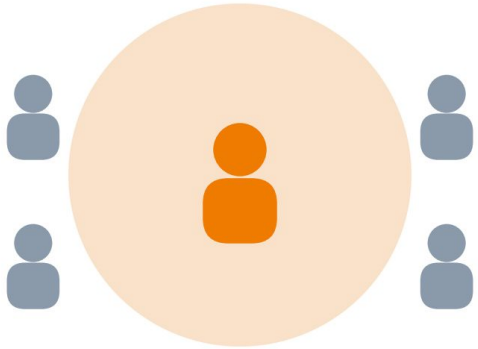
Externality: a benefit or a cost that one actor creates for others, without being paid or paying for it in the market.

Negative: a factory pollutes a river



the cost is not in the factory's price

Positive: getting vaccinated



others are protected too, without paying

Negative externality

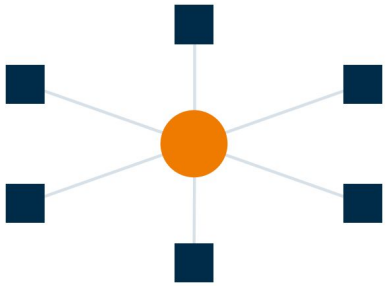
the factory earns the profits; residents pay the health and cleanup costs.

Positive externality

if you get vaccinated you also protect others, and nobody pays you for it.

Three agglomeration externalities

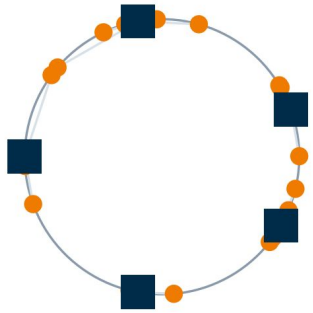
Proximity creates three positive externalities: my location choice raises the productivity of others nearby, and nobody pays for it.



1 · Input sharing

Firms close together share suppliers, infrastructure and services.

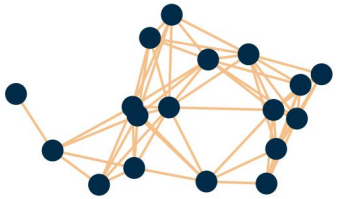
Many biotech firms in one city keep a specialised lab supplier alive.



2 · Labour market pooling

More specialised workers for firms, more employers for workers: better matches.

A software engineer has more employers to choose from in a tech cluster.



3 · Knowledge spillovers

Ideas and know-how spread when people are close.

Engineers learn from people who change jobs, from suppliers and from rivals.

Next: the third one, knowledge spillovers, is the topic of the rest of the lecture.

03

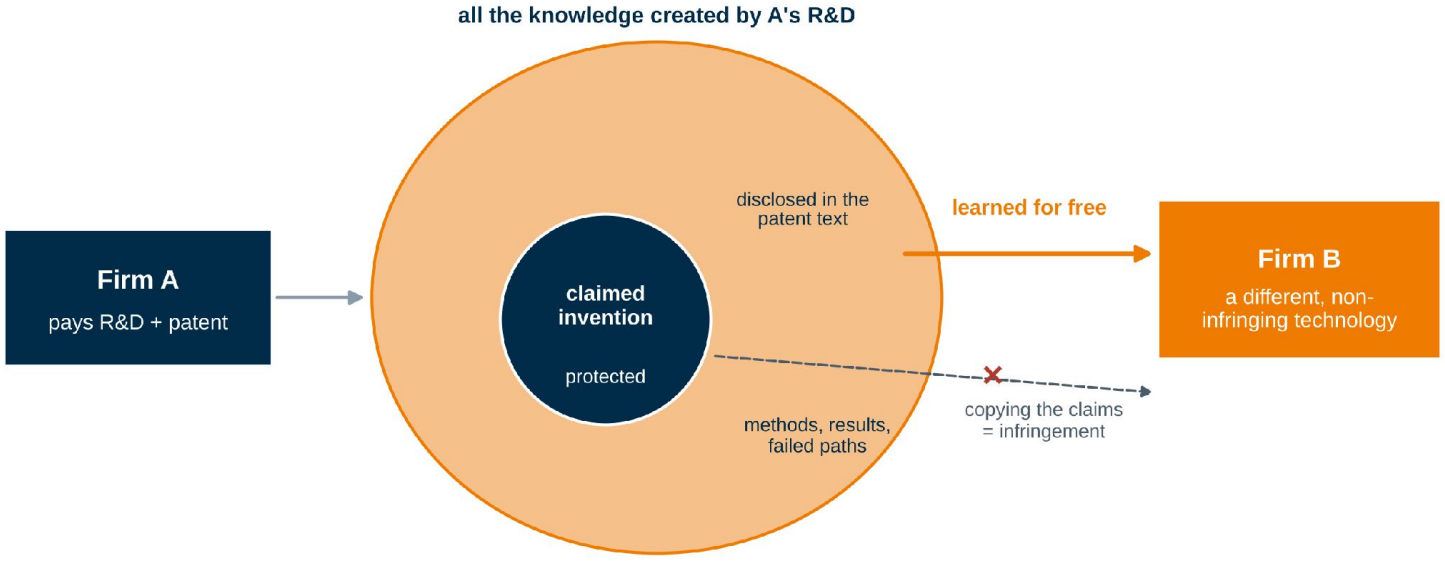
Knowledge spillovers

What they are, how they travel, why policy cares



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Why learning from a patent is a knowledge spillover



A patent protects the claimed invention, not all the knowledge behind it

1 · Firm A develops a new technology for an industrial machine and patents it.

2 · Firm A pays for the R&D and for patent protection.

3 · The patent gives Firm A exclusive rights over the claimed invention, within its legal scope and jurisdiction.

4 · Firm B learns from the technical information disclosed in the patent and develops a different, non-infringing technology.

5 · Firm B benefits from Firm A's research without necessarily paying Firm A.

The key: patents protect specific claimed inventions, not all the knowledge created during the innovation process.

Question

Is Firm B doing anything illegal? Who loses, and who gains?

Knowledge spillovers, defined

Knowledge flows: knowledge moves from A to B

paid or not

€ paid: the market works

- a licence of a patent
- a consulting contract
- buying a machine
- hiring an expert

Knowledge spillovers

not (fully) paid: an externality

- reading a published patent
- reverse engineering a product
- an engineer changes job
- a chat at a conference

Knowledge flow: any movement of knowledge from one actor to another, paid or not.

Knowledge spillover: a flow for which the originator is not (fully) paid: a positive externality.

Every spillover is a flow; not every flow is a spillover.

From Part 1: spillovers travel with people, and people meet when they are close. This is why spillovers are local.

Why knowledge spillovers matter

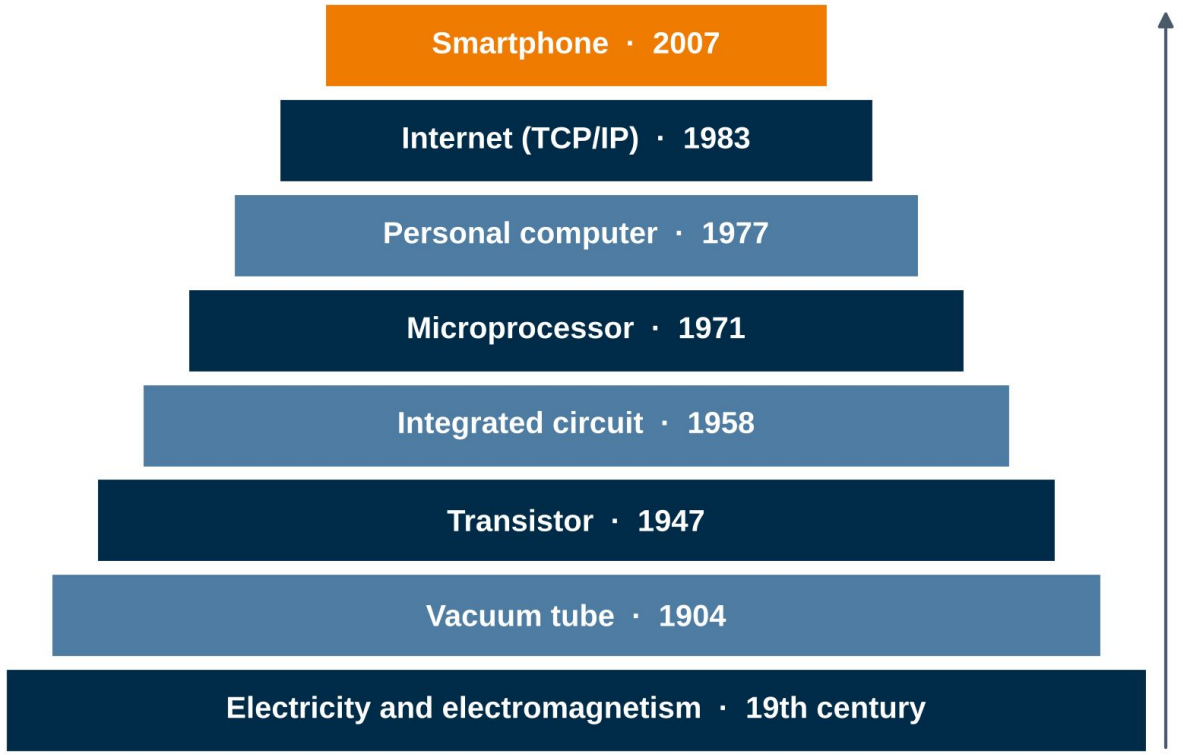
1 · Society gains more than the inventor

The social benefit of an innovation is larger than the private one. Even a patent must disclose the invention: others read it, reverse engineer the product, or reuse the idea in their own work.

2 · Innovation is cumulative

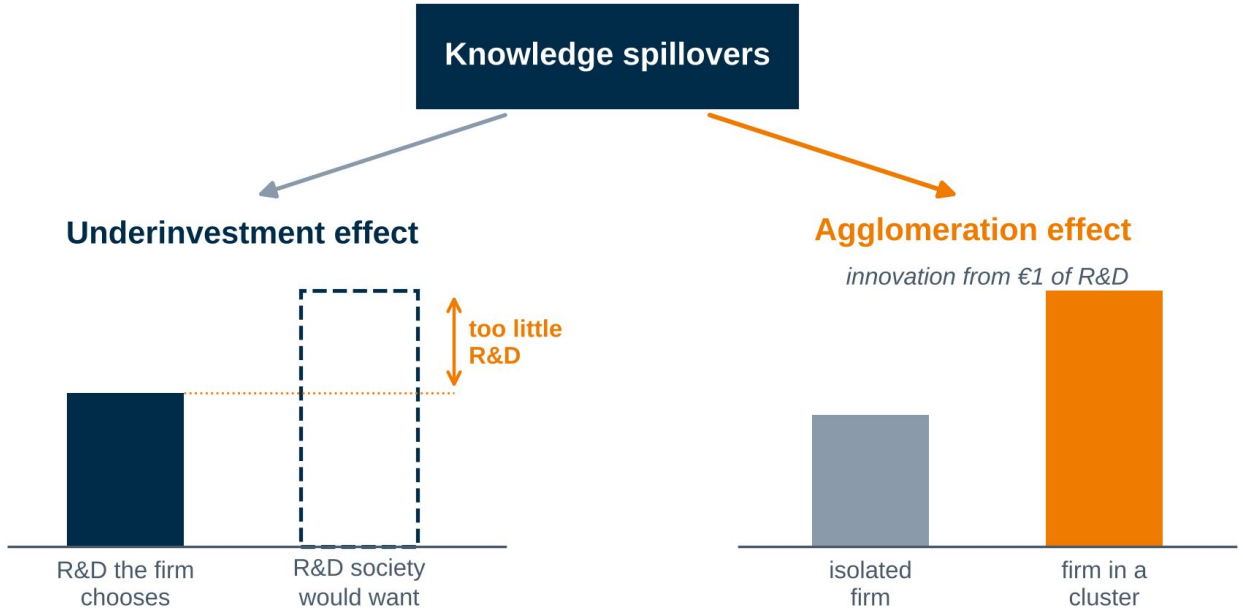
Every invention builds on earlier ones. Thanks to spillovers, each generation starts from what others already found.

> *"If I have seen further it is by standing on the shoulders of giants."*
>
> Isaac Newton, letter to Robert Hooke, 1675



Possible drawback (next slide): firms do less R&D than society would want, and this is why policy cares.

Two effects at the same time



Underinvestment effect

Firms cannot fully appropriate the social returns of their R&D, so they invest below the socially optimal level.

Agglomeration effect

Firms near other innovators benefit from external knowledge: their R&D is more productive, and this can stimulate further innovation.

Therefore: regions with strong local knowledge spillovers can show high innovation and, at the same time, socially sub-optimal R&D investment.

04

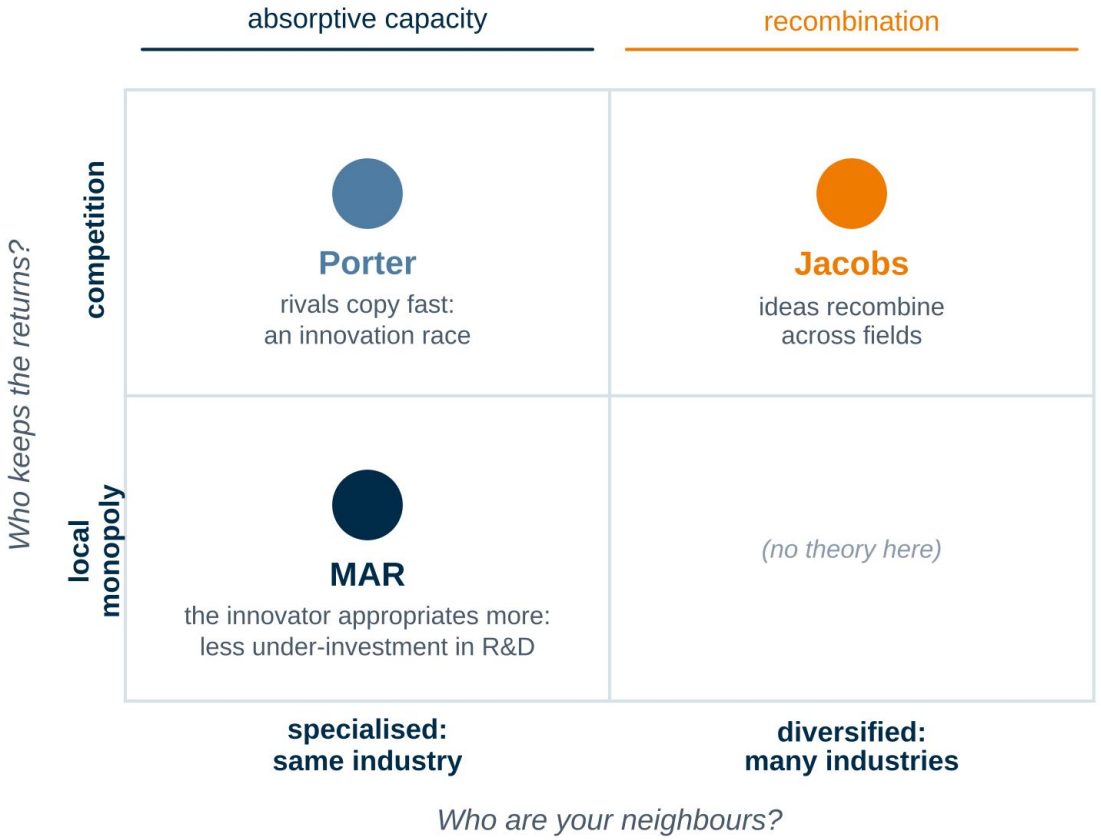
What kind of industrial environment promotes knowledge spillovers?

Marshall–Arrow–Romer, Porter, Jacobs

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How to favour knowledge spillovers? Two choices

Spillovers are good for innovation. To favour them, two elements matter: **sector specialisation** and the **appropriability regime**.



Specialised clusters (MAR, Porter)

Chips in Silicon Valley: neighbours in the same industry have the **absorptive capacity** to understand and use my knowledge.

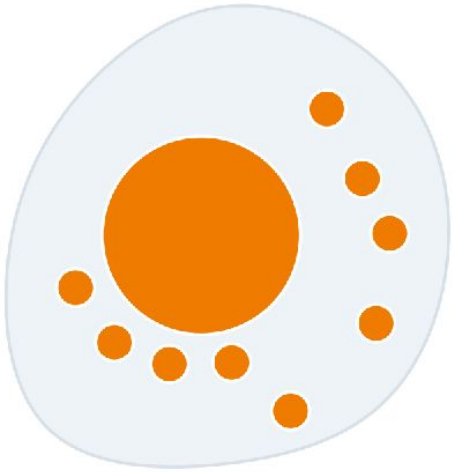
MAR: a quasi-monopoly is best: the innovator appropriates more, so there is less under-provision of R&D.

Porter: competitors are best: they start an innovation race, and everyone innovates more.

Diversified cities (Jacobs)

Incremental knowledge is cumulative: it grows inside one field. **Disruptive** knowledge is recombinant: it combines different fields. So stay close to different industries.

Cohen, W. & Levinthal, D. (1990), Absorptive capacity, Administrative Science Quarterly; Glaeser et al. (1992), Growth in cities, Journal of Political Economy.



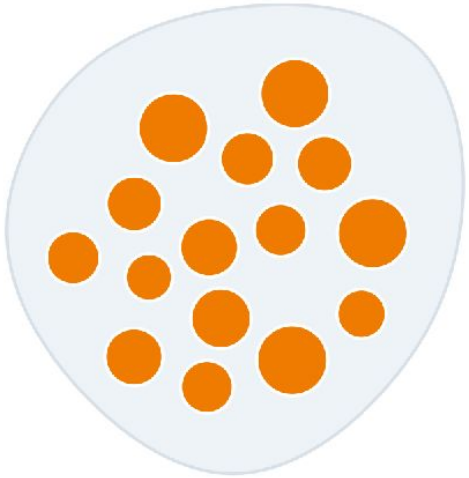
Marshall–Arrow–Romer

one industry · a dominant firm

You learn from: firms of your own industry, who can absorb your knowledge

Innovation needs: a quasi-monopoly: less under-provision of R&D

Example: Silicon Valley chips



Porter

one industry · many rivals

You learn from: firms of your own industry, who can absorb your knowledge

Innovation needs: competition: an innovation race

Example: Sassuolo ceramic tiles



Jacobs

many industries side by side

You learn from: firms of other industries

Innovation needs: variety: disruptive ideas recombine fields

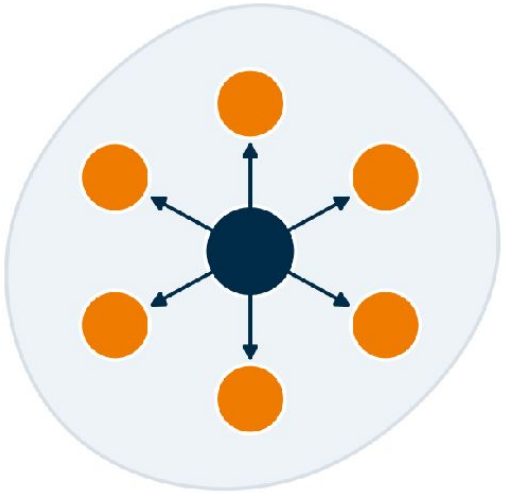
Example: Detroit: engines + carriages = cars

Two choices: specialised or diversified neighbours, and monopoly or competition (who keeps the returns of an idea).

Marshall (1890), Arrow (1962), Romer (1986); Porter (1990); Jacobs (1969).

Marshall–Arrow–Romer: learn from your own industry

Many small chip makers

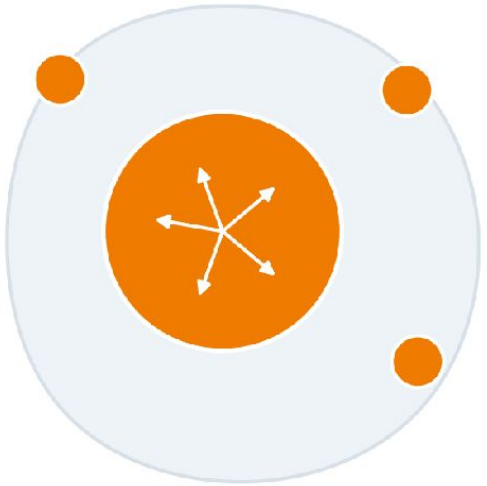


A's idea is copied at once:
A earns little, so it invests less

better for MAR

orange = chip industry
arrows = ideas that spread

One dominant chip maker



ideas circulate inside the leader:
it keeps the returns and invests more

Where knowledge comes from

other firms of the same industry nearby. They work on the same problems, so they have the absorptive capacity to understand and use your knowledge.

What helps innovation

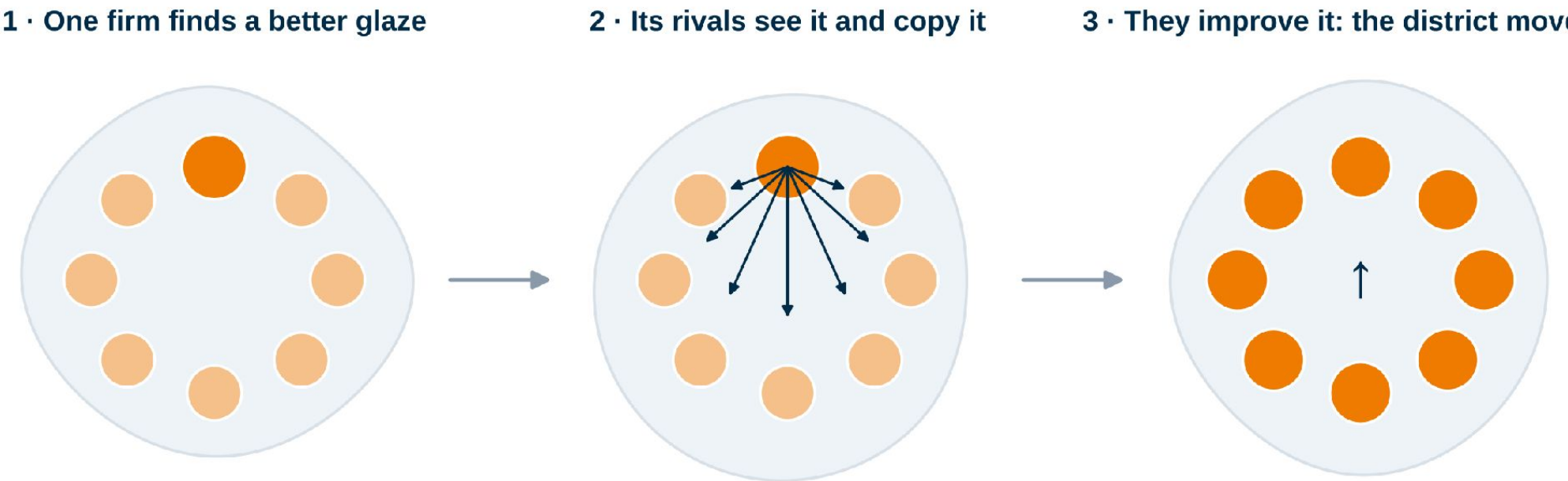
a quasi-monopoly. If ideas are copied at once, the inventor earns little and invests less (under-provision of R&D); a dominant firm keeps the returns and invests more.

Example

a specialised district: chip makers learn from other chip makers (Silicon Valley).

Marshall (1890), Arrow (1962), Romer (1986). Illustrative drawing.

Porter: learn from your rivals, compete to survive



Tile makers in Sassuolo, one industry: whoever stops improving loses its customers

Where knowledge comes from

other firms of the same industry nearby, as for MAR: specialisation gives absorptive capacity.

What helps innovation

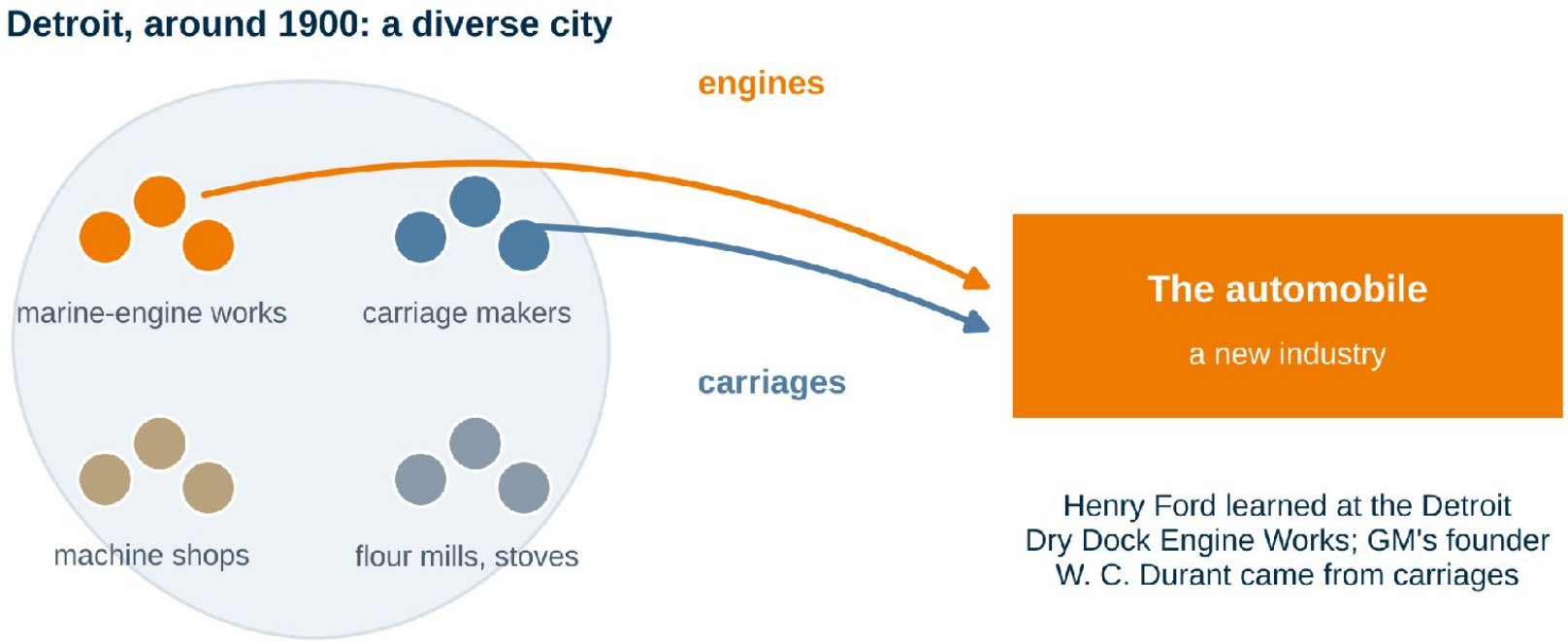
competition. Rivals copy fast, so whoever stops improving loses its customers: an innovation race pushes everyone to innovate more.

Example

the ceramic tile district of Sassuolo: a new glaze or kiln is copied, then improved by the others.

Porter, M. (1990), The Competitive Advantage of Nations. Illustrative drawing.

Jacobs: learn from other industries



A new industry from the combination of two old ones: variety lets ideas recombine

Where knowledge comes from

firms of other industries. Incremental knowledge is cumulative, inside one field; disruptive knowledge is recombinant: it combines different fields.

What helps innovation

variety: a large, diverse city where different industries meet and mix their ideas.

Example

Detroit's car industry was born from two older ones: the engine works of its shipyards and the carriage makers. Henry Ford learned at the Detroit Dry Dock Engine Works.

Jacobs, J. (1969), The Economy of Cities; Glaeser, E. (2011), Triumph of the City, and "Can Detroit find the road forward?", City Journal.

05

Tracing knowledge flows

Patent citations: a paper trail, or chasing shadows?

Politecnico di Torino

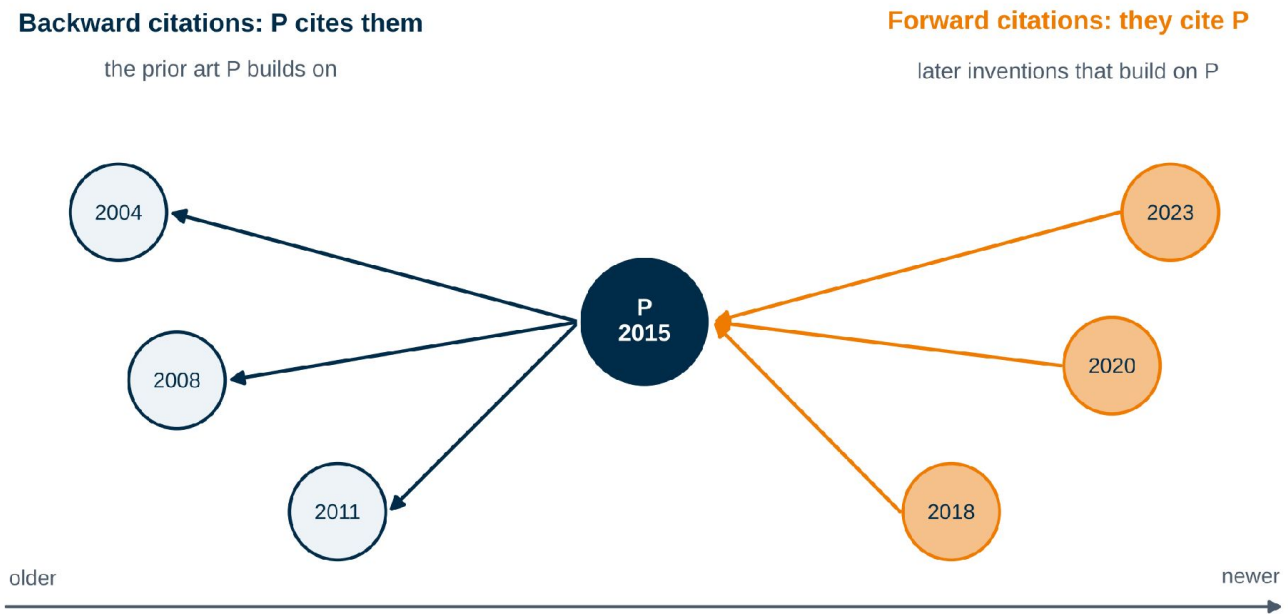
“Invisible”, or leaving a paper trail?

> “Knowledge flows … are invisible; they leave no paper trail by which they may be measured and tracked.”
>
> **Paul Krugman, Geography and Trade (1991)**

> “Knowledge flows do sometimes leave a paper trail, in the form of citations in patents.”
>
> **Jaffe, Trajtenberg & Henderson (1993)**

A patent citation points to prior art the invention builds on. If inventors cite what they learned from, citations trace knowledge flows.

Patent citations form a directed network



Nodes and links

Each patent is a node. A citation is a link from the citing patent to the cited one.

Backward citations

the patents P cites: the prior art it builds on (out-degree).

Forward citations

the later patents that cite P: inventions that build on it (in-degree).

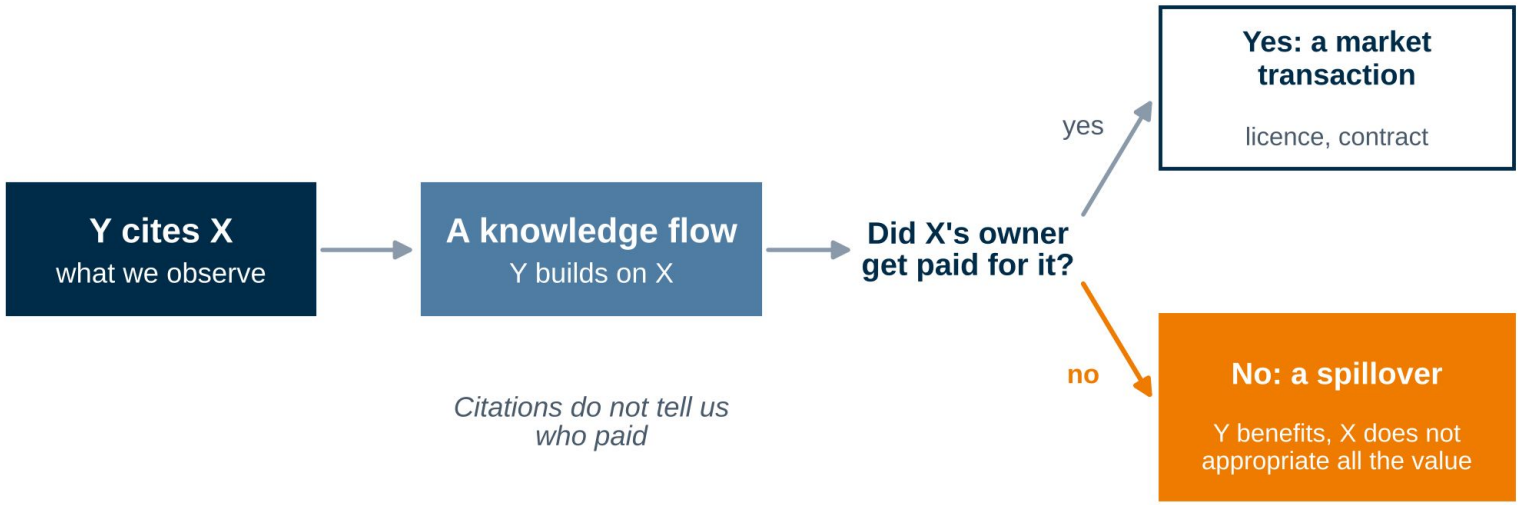
Directed

links have a direction, and point back in time: a patent can only cite older ones.

From one patent to millions: linking every patent to the ones it cites gives a network that traces how knowledge flows from one invention to the next.

We can trace knowledge flows; tracing spillovers is much harder

A knowledge flow is not, by itself, a localised externality.



To call it a spillover, two things must hold: the citing inventor benefits from the cited knowledge, and the owner of the cited patent does not appropriate all of its value.

We rarely have this information. If the owner was paid, it is not an externality but a market transaction.

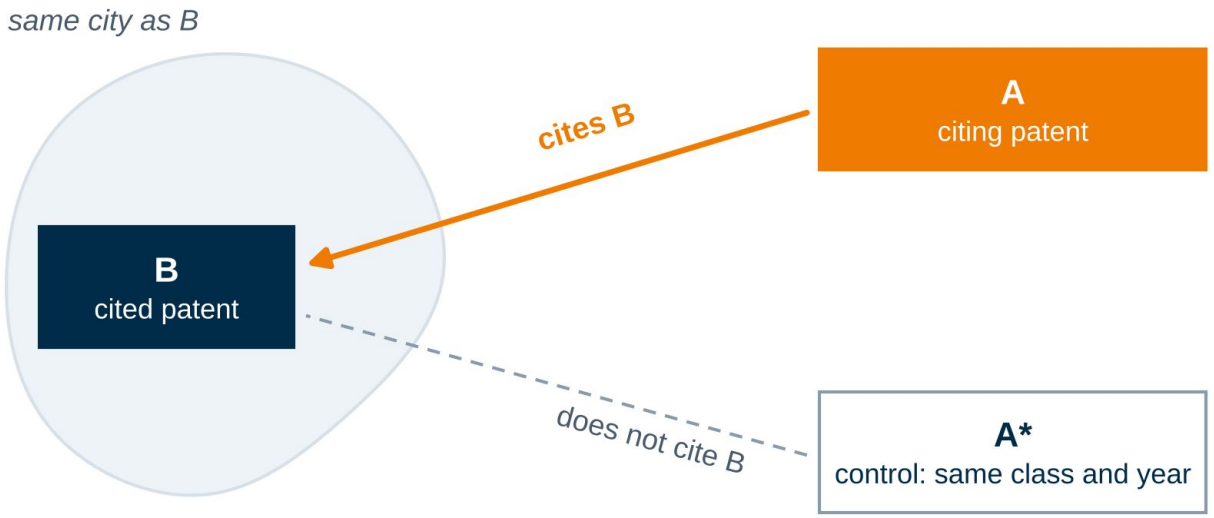
Example, labour mobility: an engineer moves to a rival with what she learned. It is a spillover: the firm she leaves gets no compensation.

The proxy is localisation: a licence works at any distance, while unpaid exchange travels with people and stays close. Citations more local than chance point to spillovers.

> **Question for you**
>
> Which other data could trace knowledge flows? Think of inventors who change job, co-authors, GitHub, the text of patents.

Jaffe, Trajtenberg & Henderson (1993); G. de Rassenfosse (2025), Patent citations: tracing spillovers or chasing shadows?

Jaffe, Trajtenberg & Henderson (1993): spillovers in local citations



Their attempt to identify spillovers

Citing patent A, cited patent B, and a control A*: same technology class and year, not citing B.

Is A in B's city more often than A*?

3× more likely in the same state; **up to 10×** in the same metropolitan area; strongest soon after B.

Self-citations are excluded: the links are between different organisations.

Question for you

Why do we need the control A*? What would we get by just counting local citations?

A fundamental paper

Jaffe, A., Trajtenberg, M. & Henderson, R. (1993), Geographic Localization of Knowledge Spillovers as Evidenced by Patent Citations, Quarterly Journal of Economics, 108(3), 577–598.

Is the trail reliable? The methodological debate

1 · How good is the control patent?

With finer technology subclasses for the controls, the localisation effect largely disappears.

2 · Who adds the citation?

Applicant or examiner: only the first can reflect what the inventor learned.

Applicant-added citations are far more localised than examiner-added ones.

Question for you

Examiners add citations to patents from anywhere in the world. If we kept them in the data, would spillovers look more local or less local than they are?

Limitations of citations as a proxy for knowledge flows

A citation should mean: “the inventor learned from the cited patent”. Two things can go wrong.

Is there a citation? \ Did knowledge really flow?	yes	no
yes	**the citation is right** *true positive*	**error of commission** *false positive*
no	**error of omission** *false negative*	**nothing to see** *true negative*

Error of commission (false positive)

There is a citation, but no knowledge flowed: the inventor never saw the cited patent, an examiner or an attorney added it. We see flows that did not happen.

Error of omission (false negative)

Knowledge flowed, but there is no citation: the inventor learned from a paper, a licence or a conversation, or chose not to cite. **Basic research**, which creates the largest spillovers, is mostly published, not patented. We miss flows that did happen.

Both errors are large: roughly **half of all citations** may not correspond to any knowledge flow.

Errors of commission: citations without knowledge flows

False positive: the citation is there, but nothing was learned.

38% of inventors knew the cited patent before or during the invention: for the others, the citation is not a flow.

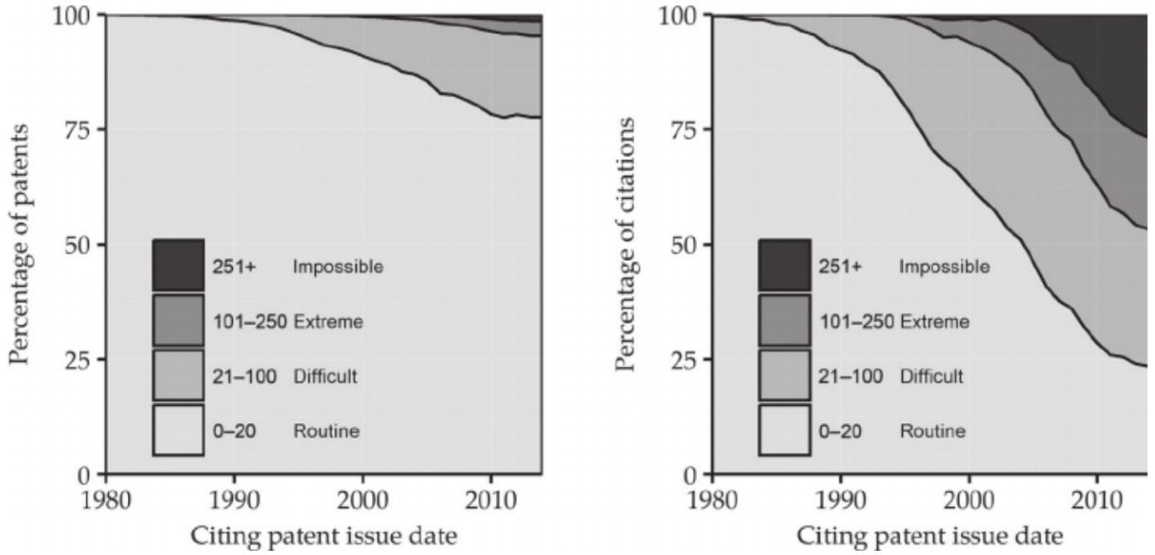
Jaffe, Trajtenberg & Fogarty (2000), survey of inventors

¼ of US citations (2005–2014) were added by the patent examiner, not by the inventor.

USPTO data, via Clancy (2021)

Attorneys too: they add citations to protect the patent in court, not because the inventor used them.

AREA PLOT OF CITATIONS BY BACKWARD CITATION COUNT OF CITING PATENT OVER TIME



(a) Percentage of patents **(b)** Percentage of citations

Super-citers: patents with more than 100 citations (darker) are under 5% of patents but about 46% of all citations: long lists added for legal safety, not learning.

So: a citation can be paperwork. Counting citations overstates knowledge flows.

Errors of omission: knowledge flows without citations

False negative: knowledge flowed, but no citation records it. The best story is the patent that founded biotechnology.

1973

Cohen (Stanford) and Boyer (UCSF) splice DNA: recombinant DNA

1974–1980

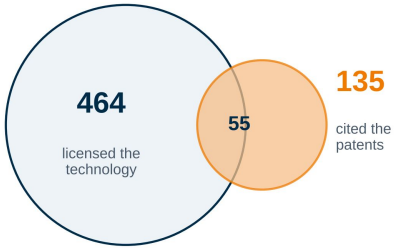
Stanford files the patents; the first is granted in 1980

1981–1997

licences to anyone who pays: 464 organisations sign one

the citation trail

only 135 organisations cite the patents; only 55 of them are licensees



What the story shows

464 organisations paid to use the technology, but only 55 of them cited the patents: citations missed 88% of the users. They learned from papers, the licence and people, not from the patent text.

Other omissions

Strategic: inventors leave out about 20% of relevant patents they know.

Basic research: university and public science create the largest spillovers, but are published, not patented: patent citations hardly see them.

Nelson, A. (2009), Research Policy; Lampe, R. (2012), REStat; Roach, M. & Cohen, W. (2013), Management Science. V

Questions?

Next PW tutorial

Knowledge spillovers · Monday 2 November, 13:00–14:30 · Room 4P + extra lecture for teamwork 17:30-19:00

Next lecture

Patent valuation · Wednesday 4 November, 13:00–14:30 · Room R3

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